School of Telecommunications Engineering
Xidian University
Xi'an 710071, China

June 29, 2026

Editor-in-Chief
Computer Networks

Dear Editor,

We are pleased to submit our manuscript entitled "NEURA: Local Excitable Routing Control for Stress-Adaptive Networks" for consideration as a research article in *Computer Networks*.

The manuscript studies adaptive routing under localized stress from a control-plane perspective. NEURA is a white-box local excitable routing-control law that combines local integration, thresholded sparse signaling, refractory suppression, short-term memory, and guarded switching to limit control dissemination, route churn, and recovery-period oscillation.

The paper evaluates NEURA against periodic traffic engineering, triggered traffic engineering, ECMP-based traffic engineering, and lightweight learning-based routing using discrete-event simulation, mechanism attribution, sensitivity analysis, safety indicators, and packet-level ns-3 validation. We believe the manuscript fits the scope of *Computer Networks* because it contributes a concrete distributed routing-control mechanism for low-overhead stress-adaptive communication systems.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests. Code, ns-3 validation programs, figure inputs, and audit outputs are available in a public GitHub reproduction repository. AI-assisted tools were used only for language editing, organization, and consistency checks; the authors reviewed and edited the content and take full responsibility for the final manuscript.

Thank you for considering our manuscript. We would be grateful for the opportunity to have it reviewed for publication in *Computer Networks*.

Sincerely,

Zhiyuan Ren
Corresponding author
School of Telecommunications
Engineering, Xidian University
Email: zyren@xidian.edu.cn